Custom Oligo Design for Target-Site Recognition Window

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Purpose

Design complementary DNA oligonucleotides to insert a target-site recognition window into the universal reporter. Prepare WT, pre-edited AC, and inactive INAC reporter sequences with the intended reading frame and directional cloning junctions.

This guide covers sequence design, checks, and synthesis specifications. Oligo annealing and reporter construction are covered in RP02.

Materials & Equipment

Item	Provider / Model	Specification / Use
Target and editor specifications	Target reference / editor design	Map the RNA strand, target C, complete recognition footprint, and spacing.
Universal reporter sequence	Full vector / cassette reference [1]	Inspect both insertion junctions and all BsaI sites.
Sequence editor	Software of choice	Reverse-complement, translation, and restriction-site checks.
Custom DNA oligos	Qualified DNA synthesis provider	Two single-stranded DNA oligos per design; 5'-phosphorylated, free 3'-OH; PAGE purification recommended for cloning. [3,4]
RNAfold (optional)	ViennaRNA Package	Brief structural screen with pairing probabilities. [7]

Before You Start

Fix the transcript, target C, and editor footprint before choosing insert boundaries. Design in the coding/sense DNA representation; write every ordered strand 5' to 3'.

Open the full construction vector. The 2,027-bp universal GenBank reference describes the cassette; additional BsaI sites in a complete plasmid must be resolved before BsaI-based RP02 construction. [1,2]

The worked example uses coding-strand fusion sequences CAGC (EGFP side) and GGCA (DHFR side). The reverse oligo starts with TGCC, the reverse complement of GGCA. Recalculate the ends if the acceptor changes. [1]

Procedure

A. Define the target window

1. Mark the target C in the reference transcript and map the complete recognition sequence and any intervening spacer. Preserve their sequence, orientation, and spacing when transferring them into the reporter. Binding-site length and editing geometry depend on the editor; they should not be inferred solely from the oligo length. [5]

2. Use the APOE4 example below as a worked design: retain 12 nucleotides immediately 5' of the target C and two nucleotides immediately 3' of it, giving a 15-nt core. The original design partitions this as the 10-nt recognition sequence ATGGAGGACG, the 2-nt spacer TG, and the target codon CGA. [1]

Element	DNA sequence (5' to 3')	Position in the core
Upstream block	ATGGAGGACGTG	1-12; recognition sequence plus spacer
Target nucleotide	C	13; first base of the target codon
Downstream bases	GA	14-15; complete the CGA codon
Complete WT core	ATGGAGGACGTGCGA	15 nt; excludes cloning overhangs

3. Make the edited base produce a stop codon in the actual reporter reading frame. In this example, the native APOE4 arginine codon CGC is synonymously changed to CGA; RNA editing then converts CGA to UGA. WT denotes the unedited reporter, whose core therefore differs from the native target at the codon's third base. Preserve the recognition footprint and flag all engineered substitutions in the design file. [1]

B. Design matched reporter controls

4. Create AC by replacing only the intended C with T in the DNA sequence. Keep the same flanking sequence, length, junctions, and backbone as WT, so AC contains the pre-edited TGA codon.

5. Create INAC with a disrupted recognition sequence and a codon that cannot become a stop codon through a single C-to-U change. In the worked example, the 15-nt INAC core is ACGTCAGTCATGGCC: its terminal GCC codon remains a sense codon after any single C-to-U change. Retain the same insert length and cloning interfaces as WT. [1]

Reporter	Core sequence (5' to 3')	Intended role
WT	ATGGAGGACGTGCGA	Unedited substrate; CGA encodes Arg and permits translation into the downstream domain.
AC	ATGGAGGACGTGTGA	Pre-edited reporter; TGA places a stop upstream of the DHFR domain.
INAC	ACGTCAGTCATGGCC	Reporter background control; both recognition sequence and terminal codon are changed.

AC provides a pre-edited reference, rather than a guaranteed fluorescence ceiling. INAC is a reporter-sequence control; a catalytically inactive editor is a separate control. Because INAC changes both recognition and codon sequence, it cannot by itself isolate the contribution of recognition-site binding.

C. Add directional cloning ends

6. For a core sequence I, write the forward oligo as CAGC + I and the reverse oligo as TGCC + reverse-complement(I), with both strands written 5' to 3'. Only the core is fully paired after annealing; the two 4-nt 5' extensions form the cloning ends. These oligos already supply the required cohesive ends and do not need added BsaI recognition sites. [1,2]

Oligo	Complete sequence (5' to 3')	Length
WT-F (APOE4-F)	CAGCATGGAGGACGTGCGA	19 nt
WT-R (APOE4-R)	TGCCTCGCACGTCCTCCAT	19 nt
AC-F	CAGCATGGAGGACGTGTGA	19 nt
AC-R	TGCCTCACACGTCCTCCAT	19 nt
INAC-F	CAGCACGTCAGTCATGGCC	19 nt
INAC-R	TGCCGGCCATGACTGACGT	19 nt

7. Reconstruct the complete assembled coding-strand junction, including vector-derived bases on both sides. For WT it is GGCAGC | ATGGAGGACGTGCGA | GGCAGC, translating to Gly-Ser | Met-Glu-Asp-Val-Arg | Gly-Ser. The overhang bases belong to these junctions and must not be counted as extra sequence added a second time. Translate across both junctions and into the downstream domain.

D. Consider flanking context and verify the design

8. For new designs, consider short transition segments on both sides of the core to reduce unfavorable pairing with adjacent reporter RNA. Keep them outside the recognition footprint, spacer, and target codon, and identical across WT, AC, and INAC. Add complete codons when the existing core starts at a codon boundary, then recheck translation. Select segments by their sequence context; adding flanks does not guarantee improved accessibility. [6]

9. Check the complete intended sequence for reverse-complement agreement, correct insertion orientation, both junctions, and the desired translation. WT and INAC must lack unintended stops before the downstream domain; AC and the simulated edited WT must stop at the intended codon. For BsaI-based construction, scan the full retained vector, insert, and new junctions for GGTCTC and GAGACC. Resolve additional sites by a compatible redesign or assembly route before construction. [2]

10. Optionally screen the assembled reporter RNA (T to U) with RNAfold at 37 C using -p -T 37. Inspect pairing probabilities over the footprint and target C, including surrounding reporter sequence; compare candidates with identical context and settings. RNAplfold -u can

additionally estimate a region's unpaired probability. Save the sequence, software version, and settings. Predictions guide design and require experimental validation. [6-8]

E. Prepare the synthesis order

11. Order each strand as single-stranded DNA with its complete 5'-to-3' sequence, including the 4-nt cloning end. Specify a 5' phosphate on both strands and an unblocked 3'-OH. If unphosphorylated oligos are selected instead, include a T4 polynucleotide kinase phosphorylation step before ligation in RP02. [3]

12. Select PAGE purification for the cloning oligos; consider HPLC when the provider recommends it for the required purity, yield, or modification compatibility. Choose a synthesis scale that supplies the planned assemblies. Save the ordered sequences, strand names, lengths, modification and purification choices, annotated assembled sequence, and the target-site mapping. [4]

p.s.

Count from the edited base, not from the first nucleotide in the recognition motif: this example has 12 bases before C and two after it. A different editor footprint may require a longer core.

An additional flanking segment belongs outside the recognition-to-target interval. Adding bases inside that interval changes editing geometry even if the overall reading frame is preserved.

Keep the 4-nt cohesive ends distinct from PCR homology tails. The reverse oligo is reverse-complemented before its TGCC end is added; reverse-complementing the entire forward oligo gives the wrong end arrangement.

Screen the complete construction vector before ordering. A clean cassette or a filename containing 'universal' does not establish that the rest of the vector is free of BsaI sites. [2]

Expected Outcome

The design package contains three matched reporter cores, six synthesis-ready DNA sequences, and an annotated assembled sequence for each variant. The example gives a 15-nt core and two 19-nt oligos per reporter, with correct directional ends and the intended WT/AC/INAC translation states.

Sequence checks establish design compatibility. Reporter construction, fluorescence behavior, and RNA editing must be assessed experimentally; a fluorescence change alone does not establish target-site editing.

Troubleshooting

Observed design problem	Check first	Action
Reverse strand does not give the intended cloning ends	Whether the entire forward strand, including CAGC, was reverse-complemented	Reverse-complement only the core, then prepend TGCC. Recheck the annealed duplex against the acceptor.
Frameshift or unintended stop in the simulated junction	Complete junction sequence, duplicated overhangs, codon boundary, and added segment lengths	Rebuild the complete coding junction and translate it. Adjust only permitted flanking bases while preserving the recognition-to-target geometry.
Editing the marked C does not produce a stop	Target base position and the actual in-frame codon	Use a compatible edit-to-stop codon; in this example, design CGA so C-to-U gives UGA. Recheck all intended sequence changes.
Restriction scan shows extra cuts	GGTCTC and GAGACC in the full retained vector, insert, and junctions	Resolve incompatible sites before RP02. Use permissible substitutions or a compatible assembly route; do not rely on the cassette-only map.
Optional RNAfold check predicts pairing over the core	Transcript context, target coordinates, and identical settings across candidates	Compare permitted external transition segments or synonymous flanking variants while preserving the core. Validate promising candidates experimentally. [6-8]

Reference

[1] SUSTech-BIO local sequence sources: REPORTER OLIGO HZ.xlsx, oligos_order_sheet-2.csv, reporter_universal.gb, and APOE4-Flag.dna. Verified 2026-10-13; full source locations and checks in source-notes.md.

[2] New England Biolabs. BsaI-HFv2: GGTCTC(1/5) cleavage and Golden Gate compatibility. Accessed 2026-10-13. Open source

[3] New England Biolabs. T4 Polynucleotide Kinase: 5' phosphorylation for oligo ligation. Accessed 2026-10-13. Open source

[4] IDT. Oligo purification methods: which type should I choose? Cloning and PAGE/HPLC; RUO23-1848_001. Accessed 2026-10-13. Open source

[5] Programmable RNA base editing with a single gRNA-free enzyme (2022). Nucleic Acids Research 50:9580-9595; CU-REWIRE geometry and PUF10. doi:10.1093/nar/gkac713. Accessed 2026-10-13. Open source

[6] Jarmoskaite et al. (2019). A Quantitative and Predictive Model for RNA Binding by Human Pumilio Proteins; scaffold effects and accessibility. Accessed 2026-10-13. Open source

[7] ViennaRNA Package. RNAfold manual, version 2.7.2; partition function (-p) and temperature (-T) options. Accessed 2026-10-13. Open source

[8] ViennaRNA Package. RNAplfold manual; -u regional unpaired probabilities and _lunp output. Accessed 2026-10-13. Open source